

Geometry Seeding

Rather than rendering at $\mathcal{F}^G(t_0)$,

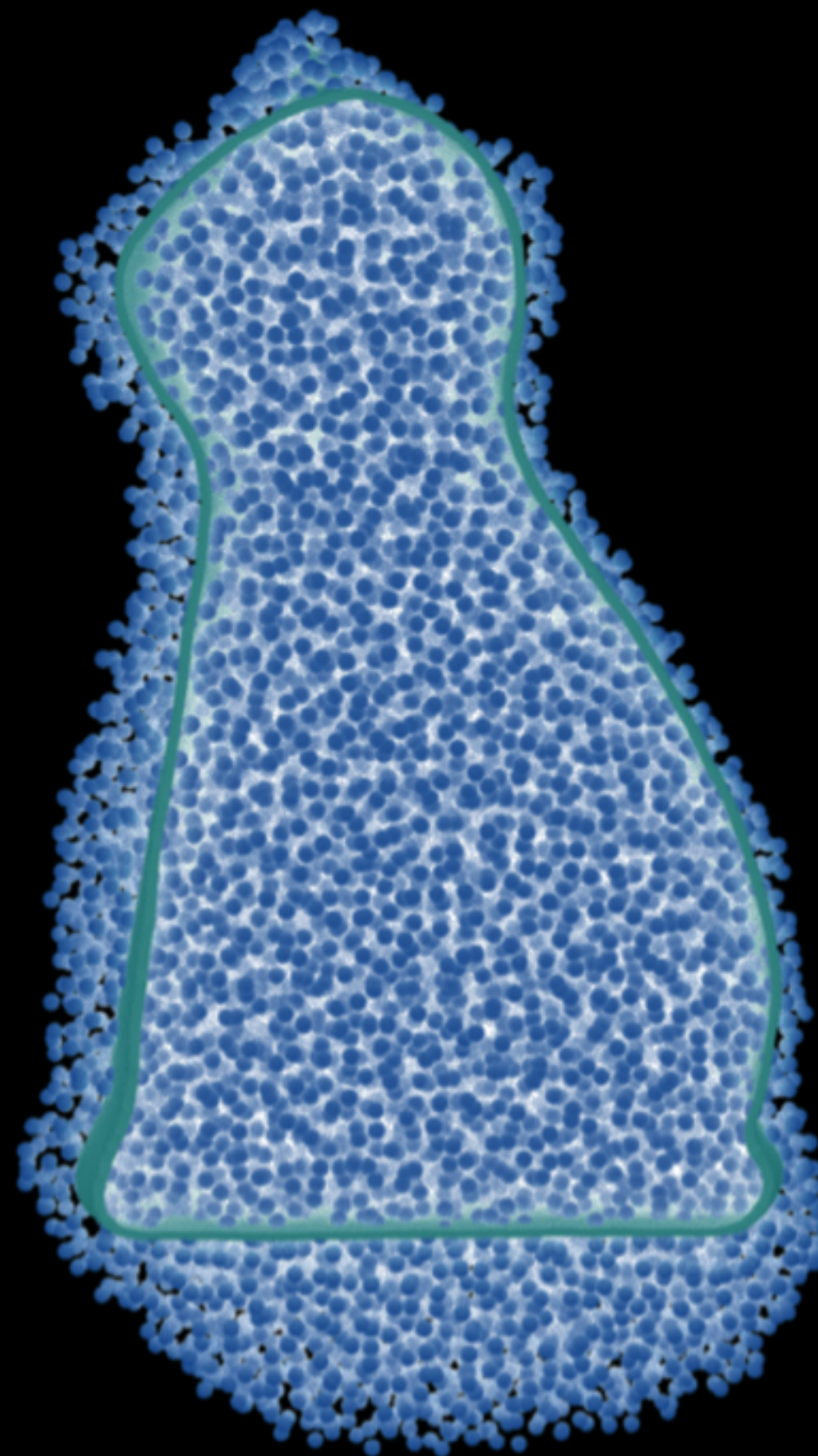
$$G2P \rightarrow P2G$$

Ensures consistent set of particles

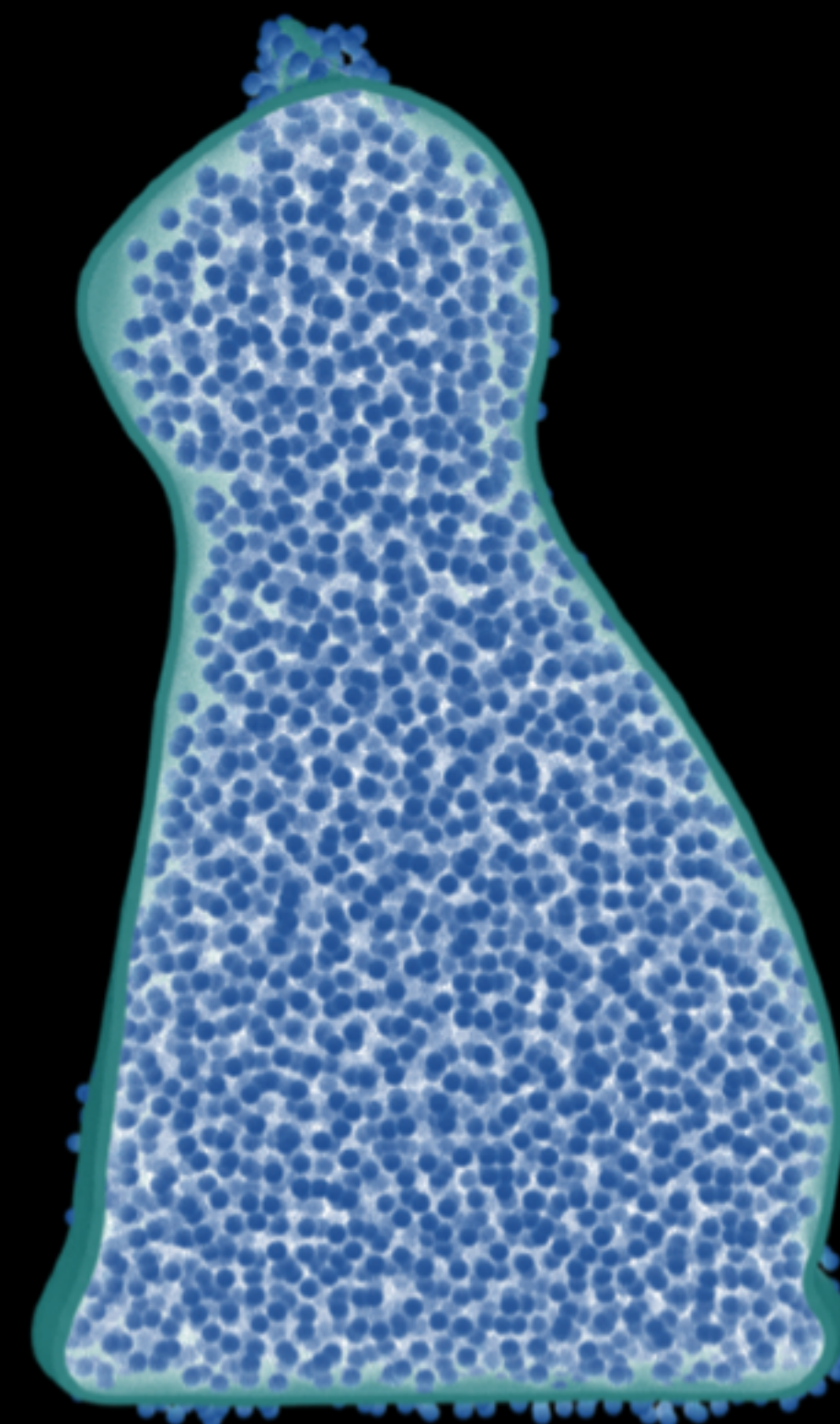
$$\mathcal{L}_{\text{surf}} = \sum_p \text{clamp}(\alpha_p, 10^{-4}, 10^{-1}) \left(\frac{\Delta x}{2} \right)^2$$

↪ Minimizes surface area

Without surface regularizer



With surface regularizer



PAC-NeRF: Physical Parameter Estimation

	Initial Guess	Optimized	Ground Truth
Droplet	$\mu = 1, \kappa = 10^3$	$\mu = 2.09 \times 10^2, \kappa = 1.08 \times 10^5$	$\mu = 200, \kappa = 10^5$
Letter	$\mu = 1, \kappa = 10^6$	$\mu = 83.85, \kappa = 1.35 \times 10^5$	$\mu = 100, \kappa = 10^5$
Cream	$\mu = 10^2, \kappa = 10^4, \tau_Y = 10, \eta = 1$	$\mu = 1.21 \times 10^5, \kappa = 1.57 \times 10^6, \tau_Y = 3.16 \times 10^3, \eta = 5.6$	$\mu = 10^4, \kappa = 10^6, \tau_Y = 3 \times 10^3, \eta = 10$
Toothpaste	$\mu = 10^2, \kappa = 10^4, \tau_Y = 10, \eta = 1$	$\mu = 6.51 \times 10^3, \kappa = 2.22 \times 10^5, \tau_Y = 228, \eta = 9.77$	$\mu = 5 \times 10^3, \kappa = 10^5, \tau_Y = 200, \eta = 10$
Torus	$E = 10^5, \nu = 0.1$	$E = 1.04 \times 10^6, \nu = 0.322$	$E = 10^6, \nu = 0.3$
Bird	$E = 10^3, \nu = 0.1$	$E = 2.78 \times 10^5, \nu = 0.273$	$E = 3 \times 10^5, \nu = 0.3$
Playdoh	$E = 10^5, \nu = 0.4, \tau_Y = 10^3$	$E = 3.84 \times 10^6, \nu = 0.272, \tau_Y = 1.69 \times 10^4$	$E = 2 \times 10^6, \nu = 0.3, \tau_Y = 1.54 \times 10^4$
Cat	$E = 10^3, \nu = 0.2, \tau_Y = 10^2$	$E = 1.61 \times 10^5, \nu = 0.293, \tau_Y = 3.57 \times 10^3$	$E = 10^6, \nu = 0.3, \tau_Y = 3.85 \times 10^3$
Trophy	$\theta_{fric}^0 = 10^\circ$	$\theta_{fric}^0 = 36.1^\circ$	$\theta_{fric}^0 = 40^\circ$

